UNIVERSITY OF MYSORE

(Re-accredited by NAAC with 'A' Grade)
(NIRF-2022: Ranked 33rd in University Category and 54th in Overall Category)

MYSORE UNIVERSITY SCHOOL OF ENGINEERING

Manasagangotri campus, Mysuru-570006
(Approved by AICTE, New Delhi)

A Project Phase-II (21CDP81) Report

On

"AI POWERED PDF QUERY SYSTEM"

Submitted in partial fulfilment for the award of the degree of
Bachelor of Engineering
in
Computer Science and Design

Submitted By

Srihari S
21SECD45

VIII Semester,
Department of CS&D
MUSE.

Under Faculty Incharge
Poornashree Narayani S Kulkarni
Asst. Professor,
Dept. of CS&D,
MUSE, UOM, Mysuru.

Head of the Department
Dr. Syed Salim
Asst. Professor,
Dept. of CS&D,
MUSE, UOM, Mysuru.

Dr. M. S. Govinde Gowda
Director,
MUSE, UOM, Mysuru.

DEPARTMENT OF COMPUTER SCIENCE AND DESIGN
MYSORE UNIVERSITY SCHOOL OF ENGINEERING
UNIVERSITY OF MYSORE
Manasagangothri campus, Mysuru-06.

UNIVERSITY OF MYSORE

MYSORE UNIVERSITY SCHOOL OF ENGINEERING

Manasagangotri campus, Mysuru-570006
(Approved by AICTE, New Delhi)

DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

CERTIFICATE

This is to certify that the **Project Phase-II (21CDP81)** title entitled as **"AI Powered PDF Query System"** is a bonafide work carried out by **Srihari S, VIII Semester** bearing **Register No. 21SECD45,** a student of **Computer Science and Design, in** partial fulfillment for the award of **Bachelor of Engineering** in the **Department of Computer Science and Design, Mysore University School of Engineering, University of Mysore, Mysuru**. It is certified that all corrections/suggestions indicated for have been executed by the above-mentioned candidate.

Signature of the Head of the Department
Dr. Syed Salim
Asst. Professor and HoD,
Dept. of CS&D,
MUSE, UOM, Mysuru**.**

Signature of Faculty Incharge,
Poornashree Narayani S Kulkarni
Asst. Professor,
Dept. of CS&D,
MUSE, UOM, Mysuru.

Signature of the Director
Dr. M. S. Govinde Gowda
Director
MUSE, UOM, Mysuru.

Name of the Examiners: **Signature with date**

1.

2.

DECLARATION

I, **Srihari S,** bearing the Register No. **21SECD45,** student of VIII semester, **Bachelor of Engineering** in the **Department of Computer Science and Design, University of Mysore, Mysuru**. Declare that the **Project Phase-II (21CDP81) entitled on "AI Powered PDF Query System**", has been duly executed by me, under faculty incharge **Poornashree Narayani S Kulkarni.**

The Project Phase-II report of the above entitled is submitted by me in the view of partial fulfillment of the requirement for the award of a Bachelor of Engineering degree in Computer Science and Design by the Department of Computer Science and Design, Mysore University School of Engineering, University of Mysore, Mysuru. During the academic year **2024-25** further, the matter embodied in the report has not been submitted previously by anybody for the award of any degree/diploma/certificate programme.

Date:15/07/2025
Place: Mysuru

Srihari S
21SECD45

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of the **Project phase -II (21CDP81)** report which would be complete only with the mention of the almighty God and the people who made it possible, whose report rewarded the effort with success of project presentation.

I am grateful to **Department of Computer Science and Design, Mysore University School of Engineering, University of Mysore, Manasagangotri campus, Mysuru** for providing me an opportunity to enhance our knowledge through the Project Phase-II.

I express my sincere thanks to **Dr. M. S. Govinde Gowda Director, Mysore University School of Engineering, University of Mysore, Manasagangotri campus, Mysuru** for providing me an opportunity, extensive support, encouragement, and means to present the Project Phase-II.

I express my heart full thanks to **Dr. Syed Salim., Head of the Department**, **Department of Computer Science and Design, Mysore University School of Engineering, University of Mysore, Manasagangotri campus, Mysuru** for encouragement in our Project Phase-II.

I would like to express profound thanks to the **Poornashree Narayani S Kulkarni, Asst. Professor**, **Department of Computer Science and Design, Mysore University School of Engineering, University of Mysore, Manasagangotri campus, Mysuru**, for the keen interest and encouragement in our Project Phase-II presentation.

Finally, I would like to thank our family members and friends for standing with me all the time.

Srihari S

21SECD45

ABSTRACT

In recent years, the need for intelligent systems that can interpret and summarize document-based information has grown significantly. This project presents the design and implementation of an AI-powered web application that allows users to interact with PDF files through a conversational interface. The system enables users to upload PDF documents, ask questions in natural language, and receive accurate responses based on the document content mimicking the capabilities of modern AI assistants like ChatGPT.

The solution adopts a modular architecture where the backend, powered by Hugging Face's transformer models, is hosted on a separate machine. This backend is responsible for parsing PDF documents, performing semantic chunking, embedding generation, and producing context-aware responses using an LLM (Large Language Model). The frontend is developed using React.js, Tailwind CSS, and Framer Motion to deliver a sleek, animated, and responsive user interface. It supports drag-and-drop PDF upload, animated typing effects, and chat-style responses to enhance user engagement.

The project leverages modern AI-driven tools like Vercel v0 for rapid UI prototyping and integrates best practices in both frontend design and backend model orchestration. The system demonstrates efficient document understanding, smooth user interaction, and scalability for larger datasets. Future enhancements include OCR support, multilingual capabilities, and deployment on cloud infrastructure for broader accessibility. This work highlights the synergy between NLP, user-centric design, and modular development in building intelligent document-interaction platforms.

Table of Contents

Sl.no	Title	Page no
1	Introduction	**1-6**
	1.1 History	1
	1.1.1 Emergence of AI-Powered Document Analysis	1
	1.1.2 Conceptualizing the Project	1
	1.1.3 Backend Development Using Hugging Face	1
	1.1.4 Enhanced Table Extraction and Structured Output	2
	1.1.5 Frontend Innovation and Interface Design	2
	1.2 Origin	
	1.2.1 Identifying the Real-World Need	3
	1.2.2 Inspiration from Generative AI Advances	3
	1.2.3 Influence of Developer-Centric Design Tools	3
	1.2.4 Parallel Development Strategy	4
	1.3 Advantages	4
	1.3.1 User-Friendly PDF Interaction	5
	1.3.2 AI-Driven Precision	5
	1.3.3 Modular Architecture	5
	1.3.4 Enhanced UI Experience	5
	1.3.5 Accelerated Development Using AI	5
	1.4 Disadvantages	6
	1.4.1 Device Dependency and Resource Consumption	6
	1.4.2 Limited Real-Time Collaboration	6
	1.4.3 Security and Privacy Concerns	6
	1.4.4 Dependency on External Libraries and Tools	6
	1.4.5 Steep Learning Curve	6
2		**7-11**
	2.1 Literature Survey	7-10
	2.2 Organisation of the Report	11
	2.2.1 Chapter-Wise Overview	11

3	Problem Statement	**12-14**
	3.1 Objectives	12
	3.2 Hardware Requirements	13
	3.3 Software Requirements	14
4	Objective – 1: Interactive Frontend with Modern UI/UX	**15-20**
	4.1 Objective	15
	4.2 Preamble	15
	4.3 Related Work	15
	4.4 Problem Statement	16
	4.5 Proposed Solution	16
	4.5.1 Dataset Details:	16
	4.5.2 Mathematical Model:	17
	4.5.3 Algorithm for File Submission	17
	4.5.4 Flowchart / Model Design / Framework	17
	4.5.5 Code Snippet (Frontend)	18
	4.5.5.1 PDF Upload Component (Simplified):	18
	4.5.5.2 Query Submission	18
		19
5	Objective – 2: Seamless Backend Integration Using Hugging Face Models	**21-29**
	5.1 Objective	21
	5.2 Preamble	21
	5.3 Related Work	22
	5.4 Problem Statement	22
	5.5 Proposed Solution	23
	5.5.1 Dataset Details	23
	5.5.2 Mathematical Model	24
	5.5.3 Algorithm	24
	5.5.4 Flowchart / Model Design / Framework	25
	5.5.5 Code Snippet	26
	5.5.5.1. File Upload Endpoint	26
	5.5.5.2. Table Extraction via Hugging Face API	27
	5.5.5.3. Smart Chunking with Table Awareness	27
	5.5.5.4. Storing Embeddings in ChromaDB	28
	5.5.5.5. Query Routing Logic	28
	5.5.5.6. LLM Answer Generation with Prompt Fallback	28
	5.5.5.7. Query Endpoint	29
	5.6 Result and Discussion	29
	5.7 Conclusion and Future Work	

6	Objective – 3: Implementing/Integrating ML Model into Web Application	**30-35**
	6.1 Objective	30
	6.2 Preamble	30
	6.3 Related Work	30
	6.4 Problem Statement	31
	6.5 Proposed Solution	31
	6.5.1 Dataset Details	32
	6.5.2 Mathematical Model	33
	6.5.3 Algorithm	33
	6.5.4 Flowchart / Model Design / Framework	34
	6.5.5 Code Snippet	35
	6.5.5.1. Query Endpoint (main.py)	35
	6.5.5.2. Prompt Routing (phi_client.py)	35
	6.5.5.3. LLM Call Using Local Ollama (phi_client.py)	35
7	Result and Discussion [Performance Evaluation Matrices]	**36-38**
	7.1Code Snippet	36
	7.2 Performance Matrices	37
	7.3 Visualization	37
	7.4 Discussions	38
	Conclusion	**39**
	8.1 Conclusion	39
	8.2 Future work	39
	References	**40**

List of Figures

Figure No.	Figure Names	Page No.
Fig 4.5.4.1	Flowchart of modal	**17**
Fig 4.5.5.2.1	Home page	**19**
Fig 5.5.4.1	Flowchart of Pdf processing	**25**
Fig 5.5.4.2	System Architecture	**25**
Fig 6.5.3.1	Relevant answering by the chatbot to queries asked about document	**34**
Fig 6.5.4.1	Flowchart of backend working	**31**